Interpretable AIGC Detection via Multi-Dimensional Linguistic Feature Engineering: A Deep Analysis of the HC3 Corpus

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ABSTRACT

The proliferation of large language models (LLMs) such as ChatGPT has raised serious concerns about academic plagiarism and misinformation. This paper addresses the problem of distinguishing human-written text from AI-generated text (AIGC detection) using the Human ChatGPT Comparison Corpus (HC3), covering both English and Chinese domains. Rather than relying on black-box deep learning models, we adopt a fully interpretable pipeline grounded in multi-dimensional linguistic feature engineering: six feature categories comprising 46–51 hand-crafted features per language (vocabulary richness, readability, POS/syntax, dependency depth, discourse patterns, and sentiment/emotion). Redundant features are removed by Pearson correlation analysis ($|r|>0.85$), and three gradient-boosted tree classifiers (Random Forest, XGBoost, LightGBM) are trained and benchmarked. Model decisions are explained via SHAP (SHapley Additive exPlanations) values and multi-model feature importance consensus. Our best models achieve **95.1%** accuracy / F1 = 0.9506 (AUC = 0.9898) on English and **92.1%** accuracy / F1 = 0.9199 (AUC = 0.9779) on Chinese, using only hand-crafted features. SHAP attribution reveals that bigram repetition, syntactic complexity, and vocabulary density are the most discriminative signals across both languages, providing clear, quantitative answers to the question: *what linguistic fingerprints expose AI identity?*

KEYWORDS

AIGC detection, feature engineering, interpretable machine learning, SHAP, HC3, ChatGPT, text mining

ACM Reference Format:
Yichong Yang, Jiahao Wei, Jiaxuan Xiong, Zizhi Zeng, and Yiming Jiang. 2026. Interpretable AIGC Detection via Multi-Dimensional Linguistic Feature Engineering: A Deep Analysis of the HC3 Corpus. In *Proceedings of Data Mining Course Final Project (CS306 DM Project).* ACM, New York, NY, USA, **??** pages.

1 INTRODUCTION

Since its public release in November 2022, ChatGPT has demonstrated the ability to produce fluent, coherent text across virtually any domain, challenging the long-standing assumption that polished writing is exclusively a human skill. While this capability unlocks tremendous productivity gains, it simultaneously introduces pressing social risks: academic fraud, large-scale generation of misinformation, and the erosion of trust in digital content [**?**].

Existing AIGC detectors largely fall into two families. *Neural detectors* fine-tune pre-trained language models (e.g., RoBERTa, DeBERTa) on labeled corpora and achieve near-ceiling accuracy but are opaque, computationally expensive, and brittle to domain shift. *Zero-shot statistical methods* (e.g., DetectGPT [**?**], GLTR [**?**]) exploit log-probability perturbations from the generating model itself, but require white-box access to the LLM.

We argue that neither family adequately serves the practitioner who needs a detector that is (a) *accurate*, (b) *interpretable*, and (c) *model-agnostic* (applicable without access to the suspected generator). Our response is a feature-engineering-first approach that systematically mines the statistical and linguistic regularities that separate human from AI writing.

Contributions.

(1) We construct a rich, *bilingual* (English + Chinese) feature space of 46–51 linguistic features organized into six interpretable categories.
(2) We demonstrate that gradient-boosted tree classifiers trained on this feature space achieve 92–95% accuracy on the HC3 benchmark, rivaling fine-tuned neural methods while remaining fully transparent.
(3) We provide the first SHAP-based attribution study on the HC3 corpus that quantitatively ranks which specific feature combinations expose AI text, and we validate the finding with multi-model consensus across three classifiers.

2 RELATED WORK

HC3 and linguistic analysis. Guo et al. [**?**] introduced the HC3 dataset and conducted linguistic comparisons showing that ChatGPT answers tend to be longer, more neutral in sentiment, and richer in discourse connectives. Their fine-tuned RoBERTa detector reached ~98.8% F1 on English but offers no interpretability.

Statistical / zero-shot detection. GLTR [**?**] visualises per-token rank distributions from a reference LM to expose machine text. DetectGPT [**?**] perturbs candidate texts and measures log-probability curvature; it needs no training data but requires LLM access.

Stylometric and feature-based detection. Stylometric approaches (Type-Token Ratio, Honore's R, readability scores) have been applied to authorship attribution for decades [**?**]. Recent work has

CS306 DM Project, Spring 2026,
2026.

adapted these to AIGC detection [?], but most studies focus on English only and do not provide SHAP attribution. Our work extends these directions to a bilingual setting with a comprehensive SHAP analysis.

3 DATASET

3.1 Human ChatGPT Comparison Corpus (HC3)

We use the HC3 dataset [?] as our *single, deep-analysed* corpus, covering both English (**HC3-EN**) and Chinese (**HC3-ZH**) subsets.

HC3-EN. collects 24 322 questions with a total of 58 546 human answers (sourced from Reddit ELI5, WikiQA, Wikipedia, medical dialogues, and financial QA) and 26 903 ChatGPT answers across five domain splits: *reddit_eli5*, *open_qa*, *wiki_csai*, *medicine*, and *finance*.

HC3-ZH. collects 12 853 questions with 22 259 human answers and 17 522 ChatGPT answers across seven domain splits: *open_qa*, *baike*, *nlpcc_dbqa*, *medicine*, *finance*, *psychology*, and *law*.

3.2 Preprocessing and Sampling

All answers shorter than 30 characters (English) or 20 characters (Chinese) are discarded to remove near-empty responses. To keep runtime tractable and class balance exact, we draw a stratified random sample of 10 000 texts (5 000 human, 5 000 AI) from each language using a fixed seed (42). The final splits are summarized in Table **??**.

Table 1: Dataset statistics after preprocessing and stratified sampling.

Split	Lang.	Human	AI	Total
Train (80%)	EN	4000	4000	8000
Test (20%)	EN	1000	1000	2000
Train (80%)	ZH	4000	4000	8000
Test (20%)	ZH	1000	1000	2000

4 FEATURE ENGINEERING

Our feature space is organized into six interpretable categories. English texts yield 51 raw features; Chinese texts yield 46, reflecting the language-specific adaptations described below.

4.1 Category 1: Vocabulary Richness

Lexical diversity measures the breadth of vocabulary employed by a writer.

- **Type-Token Ratio (TTR)**: TTR $= V/N$ where V is the number of unique tokens and N the total token count. AI text tends to have lower TTR due to formulaic phrasing.
- **Hapax Legomena Ratio**: the fraction of words appearing exactly once, reflecting authentic creative choices unique to human writers.
- **Honore's R**: $R = 100 \cdot \log N/(1 - V_1/V)$ where V_1 is the hapax count. This statistic penalises vocabulary reuse more heavily than TTR.
- **Vocabulary Density**: $V/\sqrt{N}$, a length-normalised diversity score.

For Chinese, word-level (`ttr_word`) and character-level (`ttr_char`) TTR are computed separately because Chinese word segmentation is ambiguous.

4.2 Category 2: Readability

For English we use four scores from the `textstat` library: Flesch Reading Ease, Flesch-Kincaid Grade Level, average word length (characters), and average syllables per word. For Chinese, which lacks syllabic readability formulae, we substitute proxies: average word length in characters, average sentence length in characters, and average sentence length in words.

4.3 Category 3: POS and Syntactic Features

Part-of-speech tag ratios capture grammatical style. Eight ratios are computed for English (noun, verb, adjective, adverb, conjunction, determiner, punctuation, pronoun) via NLTK's averaged perceptron tagger. Chinese POS tagging uses jieba's hidden Markov model tagger with equivalent categories (noun $n/nr/ns/nt/nz$, verb $v/vd/vn$, etc.).

Additionally, for English only, we parse each text with spaCy's dependency parser and compute five **dependency tree depth** features: maximum depth, average depth, standard deviation of depth, ratio of deep clauses (depth > 5), and average branching factor. These capture the hierarchical complexity of clause nesting, which is notably higher in AI-generated English.

4.4 Category 4: Discourse Patterns (Transition Words)

AI text is hypothesised to overuse structured discourse markers. We define 7 sub-categories of transition words (Table **??**) and compute per-category frequency normalized per 100 words. An aggregate density and a *structured marker ratio* (sequential + conclusive markers) are also computed.

Table 2: Transition word sub-categories and example markers.

Category	Examples (EN)
Sequential	firstly, secondly, finally
Conclusive	in conclusion, to summarize, overall
Contrastive	however, nevertheless, on the other hand
Additive	moreover, furthermore, in addition
Causal	therefore, consequently, thus
Exemplifying	for example, for instance, such as
Hedging	it is important to note, generally speaking

4.5 Category 5: Sentiment and Emotion

For English, VADER sentiment analysis yields compound, neutral, negative, and positive scores. An *emotion word ratio* is computed using a fixed lexicon of 18 emotionally charged words. Three *per-sentence* statistics quantify emotional variability across the text: fluctuation (standard deviation of per-sentence compound scores), range (max − min), and *flip ratio* (fraction of consecutive sentences where sentiment polarity changes). The intuition is that human writing exhibits more affective variation—the "machine flavor" of AI text is its flat, consistently neutral register.

For Chinese, VADER is unavailable; we instead use a curated positive/negative lexicon (20+18 words) and compute analogous ratio and fluctuation features, supplemented by exclamation and question mark ratios.

4.6 Category 6: Text Statistics

Length (characters, words), sentence count, average sentence length, stopword ratio, capitalization ratio (English), unique bigram ratio, digit ratio. For Chinese, an additional *four-character phrase ratio* captures usage of chengyu (classical idiom phrases), which appear more often in formal human writing.

4.7 Correlation Filtering

To mitigate multicollinearity and reduce feature dimensionality, we compute the pairwise Pearson correlation matrix on the training set and remove one feature from each pair with $|r|>0.85$, retaining the member with higher variance. This step reduces the English feature space from 51 to **42** features and the Chinese space from 46 to **40** features (Table **??**).

Table 3: Features removed by correlation filtering ($|r|>0.85$).

Language	Removed features
English (9)	`sentence_count`, `ttr`, `avg_syllables_per_word`, `flesch_kincaid_grade`, `avg_sent_len`, `sentiment_fluctuation`, `trans_sequential`, `text_len_words`, `avg_para_len`
Chinese (6)	`avg_sent_len`, `sentiment_fluctuation`, `trans_sequential`, `text_len_words`, `avg_sent_len_words`, `ttr_word`

5 EXPERIMENTAL SETUP

5.1 Models

Three ensemble tree classifiers are trained:

Random Forest (RF): 200 trees, max depth 15, minimum samples per leaf 10, trained in parallel.

XGBoost: 200 rounds, max depth 6, learning rate 0.1, subsample and column-subsample 0.8, log-loss evaluation metric.

LightGBM: 200 rounds, max depth 8, learning rate 0.1, subsample and column-subsample 0.8.

All features are standardized with zero mean and unit variance (`StandardScaler`) before training. An 80/20 stratified train-test split (random state 42) is used for all experiments. Additionally, we report 5-fold stratified cross-validation F1 on the training set using LightGBM (100 rounds) to assess stability.

5.2 Evaluation Metrics

We report Accuracy, Precision, Recall, F1-Score (macro over both classes), and ROC-AUC.

6 RESULTS

6.1 HC3 English

Table **??** reports test-set performance for all three models. XGBoost achieves the best overall result: **95.1%** accuracy, F1 = 0.9506, AUC = 0.9898. LightGBM is a close second with a 5-fold CV mean F1 of 0.9394 (±0.0040), confirming the result is stable and not a lucky train-test split.

Table 4: Test-set performance on HC3-EN (2000 samples).

Model	Acc.	Prec.	Rec.	F1	AUC
Random Forest	0.9300	0.9479	0.9100	0.9286	0.9798
XGBoost	**0.9510**	**0.9583**	0.9430	**0.9506**	**0.9898**
LightGBM	0.9490	0.9554	**0.9420**	0.9486	0.9887
5-Fold CV (LightGBM): mean F1 = 0.9394 ± 0.0040					

6.2 HC3 Chinese

Results on Chinese (Table **??**) are slightly lower, reflecting the harder linguistic disambiguation problem. LightGBM edges XGBoost: **92.1%** accuracy, F1 = 0.9199, AUC = 0.9779. The 5-fold CV mean F1 of 0.9165 (±0.0056) is equally stable.

Table 5: Test-set performance on HC3-ZH (2000 samples).

Model	Acc.	Prec.	Rec.	F1	AUC
Random Forest	0.9000	0.9049	0.8940	0.8994	0.9665
XGBoost	0.9200	0.9242	0.9150	0.9196	0.9782
LightGBM	**0.9205**	**0.9269**	0.9130	**0.9199**	0.9779
5-Fold CV (LightGBM): mean F1 = 0.9165 ± 0.0056					

6.3 Cross-Language Comparison

English is harder to fool the detector in: the XGBoost model reaches 95.1% vs. 92.1% for Chinese. One likely factor is the availability of dependency parse features (spaCy, EN only) which contribute 13.3% of total SHAP importance. Chinese compensates with richer POS taxonomy (jieba) and character-level features unavailable in English.

7 SHAP ATTRIBUTION ANALYSIS

7.1 Top Features by SHAP Value

We apply TreeExplainer [**?**] to the XGBoost model on a 500-sample subset of the test set. Table **??** and Table **??** list the top-10 features ranked by mean |SHAP| value, together with the direction of their effect on the "AI" class probability.

7.2 Feature Group Importance

Table **??** aggregates SHAP values by the six feature categories. For English, *Text Statistics* dominates (26.6%), followed by *POS & Syntax* (18.1%) and *Dependency Depth* (13.3%). For Chinese, *POS & Syntax* leads (26.7%), reflecting the greater diagnostic power of the jieba POS taxonomy.

Table 6: Top-10 SHAP features for HC3-EN (XGBoost).

Rank	Feature	SHAP	Direction
1	`unique_bigram_ratio`	1.4193	↑ in AI
2	`dep_depth_avg`	0.7910	↑ in AI
3	`hapax_ratio`	0.6410	↓ in AI
4	`adv_ratio`	0.6175	↓ in AI
5	`vocab_density`	0.5275	↓ in AI
6	`text_len_chars`	0.5098	↓ in AI
7	`stopword_ratio`	0.4575	↓ in AI
8	`sent_len_cv`	0.4020	↓ in AI
9	`punct_ratio`	0.3110	↓ in AI
10	`paragraph_count`	0.2871	↑ in AI

Table 7: Top-10 SHAP features for HC3-ZH (XGBoost).

Rank	Feature	SHAP	Direction
1	`unique_bigram_ratio`	0.7501	↑ in AI
2	`vocab_density`	0.6175	↓ in AI
3	`adv_ratio_zh`	0.6094	↓ in AI
4	`pronoun_ratio_zh`	0.5669	↑ in AI
5	`sent_len_cv`	0.5117	↓ in AI
6	`conj_ratio_zh`	0.4234	↑ in AI
7	`honore_r`	0.4022	↓ in AI
8	`trans_hedging`	0.3764	↑ in AI
9	`avg_para_len`	0.3322	↑ in AI
10	`avg_word_len_chars`	0.3258	↓ in AI

Table 8: Feature group SHAP contribution.

Feature Group	EN (%)	ZH (%)
Text Statistics	26.6	19.1
POS & Syntax	18.1	26.7
Dependency Depth	13.3	—
Vocabulary Richness	11.3	15.9
Sentence Structure	9.8	18.0
Transition Words	8.8	8.5
Sentiment & Emotion	6.9	5.0
Readability	5.2	6.8

7.3 Multi-Model Feature Importance Consensus

To verify that the SHAP ranking is not model-specific, we rank features by impurity-based importance in all three models and compute a consensus rank (sum of ranks; lower is better). Table **??** shows the top-7 consensus features for English. All three models unanimously rank `unique_bigram_ratio` first (), strongly validating its diagnostic power.

8 KEY FINDINGS AND DISCUSSION

Finding 1: Bigram repetition is the universal AI fingerprint. `unique_bigram_ratio` (higher in AI) ranks first by SHAP in both languages and is the only consensus feature in both English and Chinese models. ChatGPT generates text with more repetitive, formulaic word-pair combinations than human writers across all domains.

Table 9: Multi-model consensus (HC3-EN). : top-tier consensus.

Feature	RF	XGB	LGB	Tier
`unique_bigram_ratio`	1	1	1	
`dep_depth_avg`	3	2	4	
`text_len_chars`	5	6	3	
`hapax_ratio`	2	3	9	
`vocab_density`	7	8	6	
`sent_len_cv`	10	9	2	
`adv_ratio`	8	12	11	

Finding 2: Syntactic complexity distinguishes AI in English. Average dependency tree depth (`dep_depth_avg`) is the #2 feature in English, contributing 13.3% of group SHAP. AI-generated English text features significantly deeper clause hierarchies—subordinate clauses nested within subordinate clauses—a pattern consistent with ChatGPT's instruction-following optimization for completeness.

Finding 3: POS profile drives Chinese detection. In Chinese, POS & Syntax features collectively account for 26.7% of total SHAP. Humans use more adverbs (`adv_ratio_zh` ↓ in AI), reflecting colloquial hedging and emotional intensification, while AI overuses conjunctions (`conj_ratio_zh` ↑ in AI) to enforce explicit logical structure.

Finding 4: Vocabulary diversity is consistently lower in AI text. Both `hapax_ratio` and `vocab_density` are negatively associated with AI text across both languages. Despite ChatGPT's vast training vocabulary, its output reuses a more constrained lexical set within any given response, possibly due to next-token prediction converging to high-probability, domain-common words.

Finding 5: Human writing shows more sentence-length variability. `sent_len_cv` (coefficient of variation of sentence lengths) is lower in AI text in both languages, indicating that ChatGPT tends to produce sentences of more uniform length—another manifestation of its "machine-flavored" regularity.

Finding 6: Transition/discourse features confirm AI's structured rhetoric. Hedging expressions (`trans_hedging`) rank #8 in Chinese SHAP, confirming that phrases like "" ("it is worth noting") and "" ("generally speaking") are disproportionately frequent in AI text. Collectively, transition word features contribute 8.5–8.8% of total SHAP, a modest but consistent contribution.

Sentiment flatness: a removed-but-valid signal. `sentiment_fluctuation` was removed by correlation filtering in both languages because it correlates ($|r|$>0.85) with other sentiment statistics. However, `sentiment_neutrality` (higher in AI) survives and contributes to the Sentiment & Emotion group. This supports the hypothesis that AI text lacks genuine emotional variation—the well-documented "machine flavor" of tonally flat prose.

Comparison to neural baselines. Our 92–95% accuracy using only hand-crafted features compares favorably to the HC3 baseline detectors: Guo et al. report ∼98.8% F1 with fine-tuned RoBERTa [**?**], a 3–7% gap that is the price of full interpretability. Unlike RoBERTa, our model explains every prediction through SHAP values, identifies the specific linguistic features driving each decision, and requires no GPU infrastructure at inference time.

9 CONCLUSION

We presented an interpretable, feature-engineering-first pipeline for bilingual AIGC detection using the HC3 corpus. Six categories of 40–42 hand-crafted linguistic features—spanning vocabulary richness, readability, syntactic structure, discourse patterns, and sentiment—feed three gradient-boosted tree classifiers (RF, XGBoost, LightGBM). SHAP attribution quantitatively reveals that *bigram repetition, syntactic clause depth, and vocabulary diversity* are the dominant signals exposing AI text, with consistent cross-language and cross-model agreement.

Our work demonstrates that accurate, interpretable AIGC detection is achievable without black-box deep learning. The feature taxonomy and SHAP methodology developed here can readily be extended to new languages, new LLMs, and domain-specific detection scenarios.

Limitations and Future Work. The current feature set does not model *topic or domain distribution* shifts; future work should evaluate cross-domain generalization. Dependency parsing is available for English only; adding a Chinese dependency parser (e.g., LTP) could narrow the 3% accuracy gap. As LLMs evolve, the feature thresholds may drift; periodic recalibration will be needed.

ACKNOWLEDGMENTS

We thank the HC3 dataset contributors (Guo et al.) for making their corpus publicly available, enabling rigorous reproducible research in AIGC detection.

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